TECHNICAL REPORT

VLMProbe:

An Open-source Library for VLM Probing and Interpretation

Exploring Model Internals Beyond the Answer

Gexin Huang[1,2], Haisheng Fu[1,2], Yanting Yang[1,2], Ruinan Jin[1,2], Jiaqi Liu[1,2], Xiaoxiao Li[1,2]

[1]The University of British Columbia [2]Vector Institute

ABSTRACT

vlmprobe brings lens methods, attention attribution, and causal interventions into a common interface for pretrained vision-language models. This report demonstrates the library through matched case studies: the same input is examined with several methods, and differences are interpreted in terms of the question each method answers. Five VLMs are evaluated on examples from GQA, MMVP, VSR, and VisOnlyQA; independently calibrated readouts and paper-aligned GPT-2 and ViT studies extend the comparison. Residual and head lenses reveal different stages and components of an answer. Attention explanations differ substantially under alternative propagation rules, and a zero local map can coexist with measurable visual dependence. Actual patching also reveals failures of first-order attribution estimates under large replacements. The library makes these comparisons explicit by retaining inputs, activation sites, target scores, and measured effects. The results illustrate analytical capabilities and method boundaries through individual cases rather than dataset-level performance rankings.

1 Introduction

An intermediate vocabulary prediction, a bright image heatmap, and a large patching effect can all appear to explain a model's answer. They provide different evidence. A lens decodes a representation, attention attribution assigns relevance under a chosen rule, and an intervention measures what changes when a specified computation is altered. Comparing these methods requires more than placing their figures side by side: the input, target, analysis unit, and intervention must be clear.

vlmprobe provides a shared interface for making such comparisons. Its adapters expose model-specific activation sites and token layouts, while method classes implement reusable readout, attribution, and intervention algorithms. Users bind an existing model once and configure tools through `probe.lens`, `probe.attention`, and `probe.causal`. The library also supplies fitting utilities for methods that require calibration and visualizations that retain the input and the measured quantity.

This report asks what the library reveals when several tools examine the same decision. Within each family, we compare methods on matched examples and explain their strengths and limitations immediately after the results. Methods with different original inputs or computational requirements receive separate, paper-aligned studies. Together, these studies show how the library supports reusable analysis across models without conflating a decoded word, an attribution score, and a causal effect.

2 Library and evaluation protocol

2.1 Model adapters and method configuration

Figure 1 summarizes the library workflow. An input contains the native model tensors and their text/image coordinates. Configuration specifies the method, activation sites, positions, and target score; paired inputs and calibrated artifacts are supplied where required. Outputs retain these choices alongside decoded tokens, attribution maps, or intervention effects. This shared contract allows methods to reuse an input preparation step and a visualization without assuming identical internal architectures.

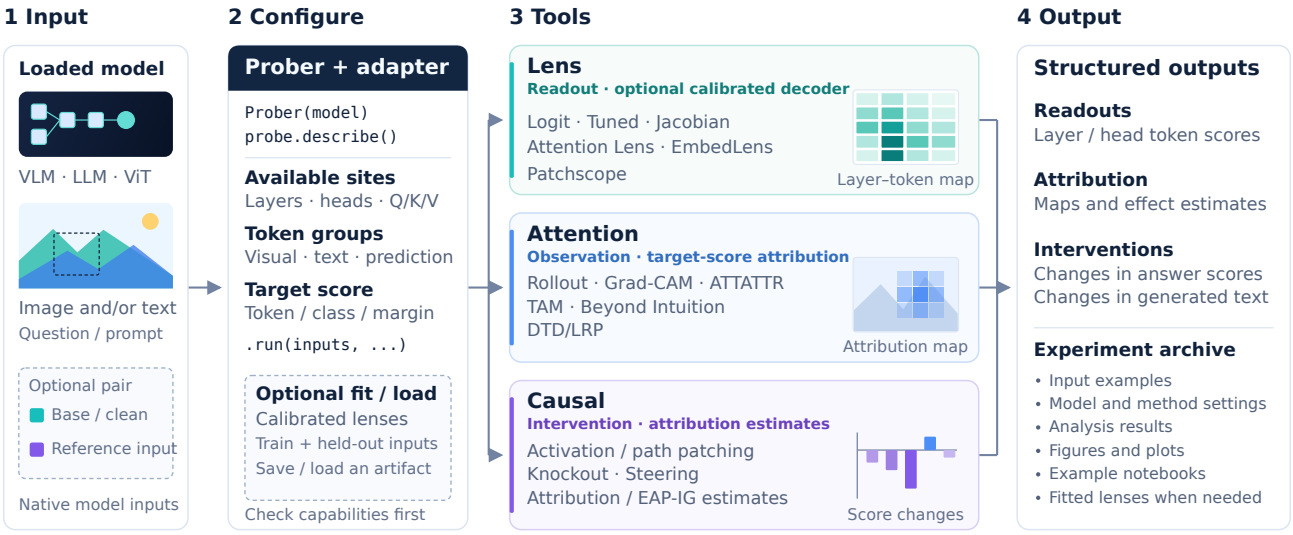


Figure 1. The vlmprobe workflow. Native inputs and explicit analysis settings feed three method families; their outputs support token readouts, spatial explanations, and measured intervention comparisons. Miniature plots illustrate output types rather than experimental results.

The following example reads a representation, explains an answer with attention gradients, and compares exact and first-order residual replacement:

```
from vlmprobe import Prober
from vlmprobe.metrics import SequenceLogProb

probe = Prober(model, processor)
metric = SequenceLogProb(answer_mask)

readout = probe.lens.logit(
    layers=layers, tokens=prediction_positions
).run(inputs)
attention = probe.attention.grad_cam(
    layers=attention_layers, queries=prediction_mask, keys="visual"
).run(inputs, metric=metric)
exact = probe.causal.patch(layers=layers, tokens="visual").run(
    inputs, source=reference, metric=metric
)
estimate = probe.causal.attribute(layers=layers, tokens="visual").run(
    inputs, source=reference, metric=metric
)
```

Here `inputs` includes the fixed answer for teacher-forced scoring; `answer_mask` selects its token pieces, and `reference` supplies aligned source activations. The method implementations share the adapter interface, while `probe.describe()` exposes capabilities relevant to choosing sites and tools. The executed notebooks supply the native preprocessing and position-selection steps omitted from this compact example.

2.2 Models, inputs, and comparison groups

Table 1 lists the five pretrained VLMs used for the shared case studies. They vary in decoder architecture, visual-token compression, and model size. Table 2 defines the eight fixed inputs, chosen before inference. GQA supplies a common attribute question and target boxes (Hudson and Manning 2019); MMVP contrasts visually similar images (Tong et al. 2024); VSR tests spatial propositions (Liu et al. 2023); and VisOnlyQA supplies diagrams and charts (Kamoi et al. 2025).

Table 1. Models in the shared-case comparison. Sizes follow public checkpoint names. Native adapters retain each model's decoder sites and visual-token grid.

Model	Image grid	Architectural feature
SmolVLM-500M-Instruct	8×8	Compressed image tokens with a Llama decoder
InternVL3-1B-hf	16×16	InternVL vision path with a Qwen2 decoder
Qwen3-VL-2B-Instruct	14×14	Normalized Q/K and DeepStack visual additions
Pixtral-12B	16×16	Pixtral patch encoder with a Mistral decoder
LLaVA-1.5-7B	24×24	CLIP visual features projected into a Llama decoder

Table 2. Predetermined shared inputs. Each source contributes two cases; image captions from a separate COCO split calibrate the VLM Tuned Lenses.

Source	Released identifiers	Analysis question
GQA	07302654 / 19225230	Is the shirt black or yellow?
MMVP	Images 1 / 2	Are the butterfly's wings open or closed?
VSR zero-shot	Test rows 0 / 1	Car behind suitcase; cat under backpack
VisOnlyQA real	Row 0 of two splits	Connected triangle; chart-value extraction

The same photograph and semantic question are supplied to each VLM. Native chat templates and tokenizers remain model-specific. Images are padded to a square before preprocessing; displayed maps are transformed back to the original photograph, including its aspect ratio and bounding-box coordinates. Comparisons within a model hold token positions fixed. Comparisons across models preserve semantic targets, since native subwords and visual grids differ. The generation-steering comparison separately uses COCO validation image 310196 across three VLMs, retaining the original image geometry (Figure 13).

For a fixed answer $a = (a_1, \dots, a_m)$, the shared score is

$$S(x, a) = \sum_{j=1}^{m} \log P(a_j \mid x, a_{<j}), \qquad \Delta S = S(x \text{ with intervention}, a) - S(x, a).$$

The score includes every native answer piece and excludes EOS. A negative ΔS means reduced support for that answer. Native generation is shown separately so that an explanation of a forced answer is distinguishable from an explanation of the model's actual response. Different tokenizations make absolute scores unsuitable as a cross-model calibration ranking.

The comparison groups also depend on the method's analysis unit. Table 3 outlines these groups. GPT-2 and ViT studies retain the original text and vision settings for tools whose calibrated artifacts or audited computational graphs are not yet available on all five VLMs. These studies complement the shared cases rather than forming one universal method leaderboard. Training, checkpoint revisions, literal prompts, and validation checks are provided in Appendix A.

The comparisons cover the paper-linked analysis and intervention methods. Attention profiling, head-logit inspection, temperature scaling, and reweighting are also available as configurable utilities; they are not counted as separate paper-reproduction studies.

Table 3. Comparison groups and the evidence each provides. Groups organize matched inputs and compatible measures; distinct analysis units remain separately interpreted.

Family	Methods demonstrated	Matched comparison
Lens	Logit, Tuned, Jacobian, Attention Lens; EmbedLens readout; Patchscope	Residual/head decoding on IOI; visual-token and transfer readouts on GQA
Attention	Rollout, Grad-CAM, ATTATTR, TAM, Beyond-H/C, generic relevance	Seven attribution methods on GQA across five VLMs; native ViT maps and deletion tests
Causal	Activation/attribution patching, knockout, path patching, EAP/EAP-IG, VSV steering	Exact versus approximate replacements; attention knockout; controlled sender paths; edge retention; generation changes across three VLMs

3 Lens methods

3.1 Residual and head readouts expose different levels of a decision

Logit Lens applies the model's final normalization and unembedding to an intermediate residual (nostalgebraist 2020). Tuned Lens first applies a fitted affine translator, aligning that representation with the final predictive basis (Belrose et al. 2023). At layer ℓ and position t,

$$p^{\text{logit}}_{\ell,t} = \text{softmax}(W_U \,\text{Norm}(h_{\ell,t})), \quad p^{\text{tuned}}_{\ell,t} = \text{softmax}(W_U \,\text{Norm}(A_\ell h_{\ell,t} + b_\ell)).$$

Figure 2 shows the same yellow-shirt GQA question across five VLMs. Rows are decoder layers; columns are selected native prompt and answer positions. Cell text gives the most likely vocabulary fragment after native tokenizer decoding, while color gives the rank of the fixed next token. The top-1 search covers the full vocabulary without restricting the language, so English inputs can yield non-English fragments. These are intermediate vocabulary projections, not generated answers or evidence of a particular internal reasoning language. Most context columns are native chat-template suffixes, rather than aligned question words across models. The outlined answer columns expose a concrete disagreement: Qwen3-VL ranks Yellow second at the final readout and generates Black; the other models generate Yellow. Early decoded fragments describe an evolving vocabulary projection, but cannot by themselves establish that the shirt was recognized or locate a causal loss of visual information.

Two other lenses change the readout question. Jacobian Lens uses a calibrated causal transport to the final residual basis to expose verbalizable content (Gurnee et al. 2026). Attention Lens learns a vocabulary decoder for an individual head's projected output (Sakarvadia et al. 2023). These are different units and objectives: head readouts resolve components combined in a residual, whereas Jacobian readouts need not represent the next-token predictive distribution.

We examine all four lenses on a GPT-2 IOI prompt ending "Andrew gave a computer to," whose correct next name is Danielle. The native next-token probabilities are 0.842 for Danielle and 0.003 for Andrew. Figure 3 holds the final prediction position and target token fixed. Logit and Tuned Lens first select Danielle at layer 9. At layer 7, calibration improves her rank from 1,880 to 897, while Jacobian Lens ranks her 457th and instead decodes "kids." At layer 8, Jacobian Lens places Danielle at rank 35, compared with 482 for Logit Lens and 365 for Tuned Lens. All three select Danielle from layer 9. Attention Lens reveals variation within layer 10: heads 0, 2, 6, and 10 decode Danielle, while the other eight do not.

This distinction is useful when selecting subsequent interventions. A residual lens traces how a prediction emerges; a head lens nominates finer components for testing. A decoded name alone does not establish that a head is necessary or identify its functional role. Comparing calibrated head and residual probabilities as if they were additive causal contributions would obscure this difference.

Tuned Lens is calibrated against the final next-token distribution, whereas Jacobian Lens estimates downstream causal transport and Attention Lens decodes individual head messages. Those objectives do not

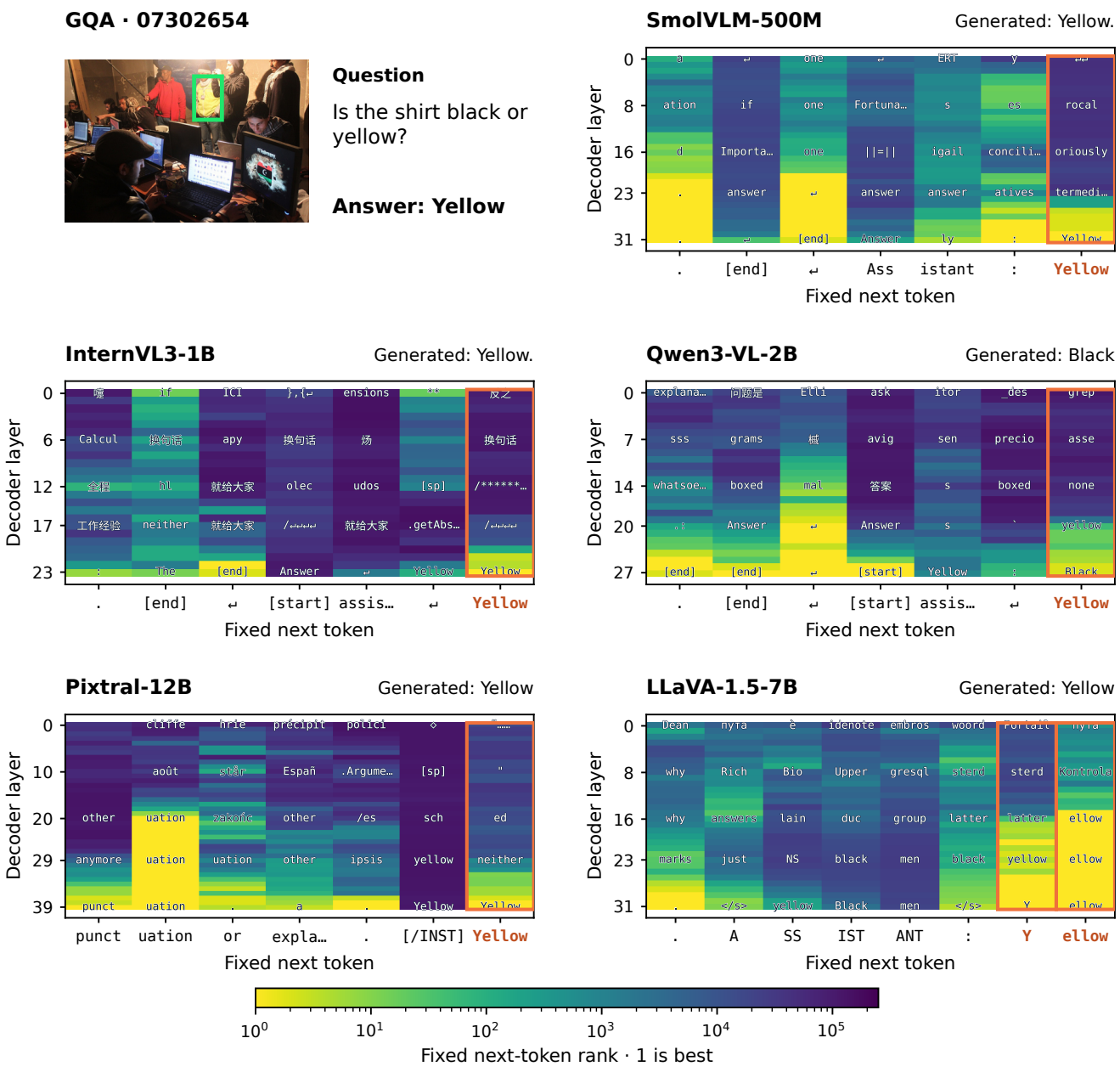


Figure 2. Layer-wise vocabulary readouts for one GQA decision across five VLMs. The input panel shows the question, answer and annotated shirt. All decoder layers retain their measured colors; selected rows show shortened top-1 fragments after native tokenizer decoding. Top-1 search includes all vocabulary languages, regardless of the English input. Undecodable or unrenderable fragments remain unlabeled. Yellow indicates a better rank for the fixed next token, using one shared logarithmic scale. Orange outlines identify answer predictions: Qwen3-VL generates Black while the other models generate Yellow. LLaVA predicts Yellow in two pieces, with the second conditioned on the first. Horizontal labels shorten chat markers to [start]/[end], omit whitespace markers and display line breaks as arrows; columns follow each model's native template.

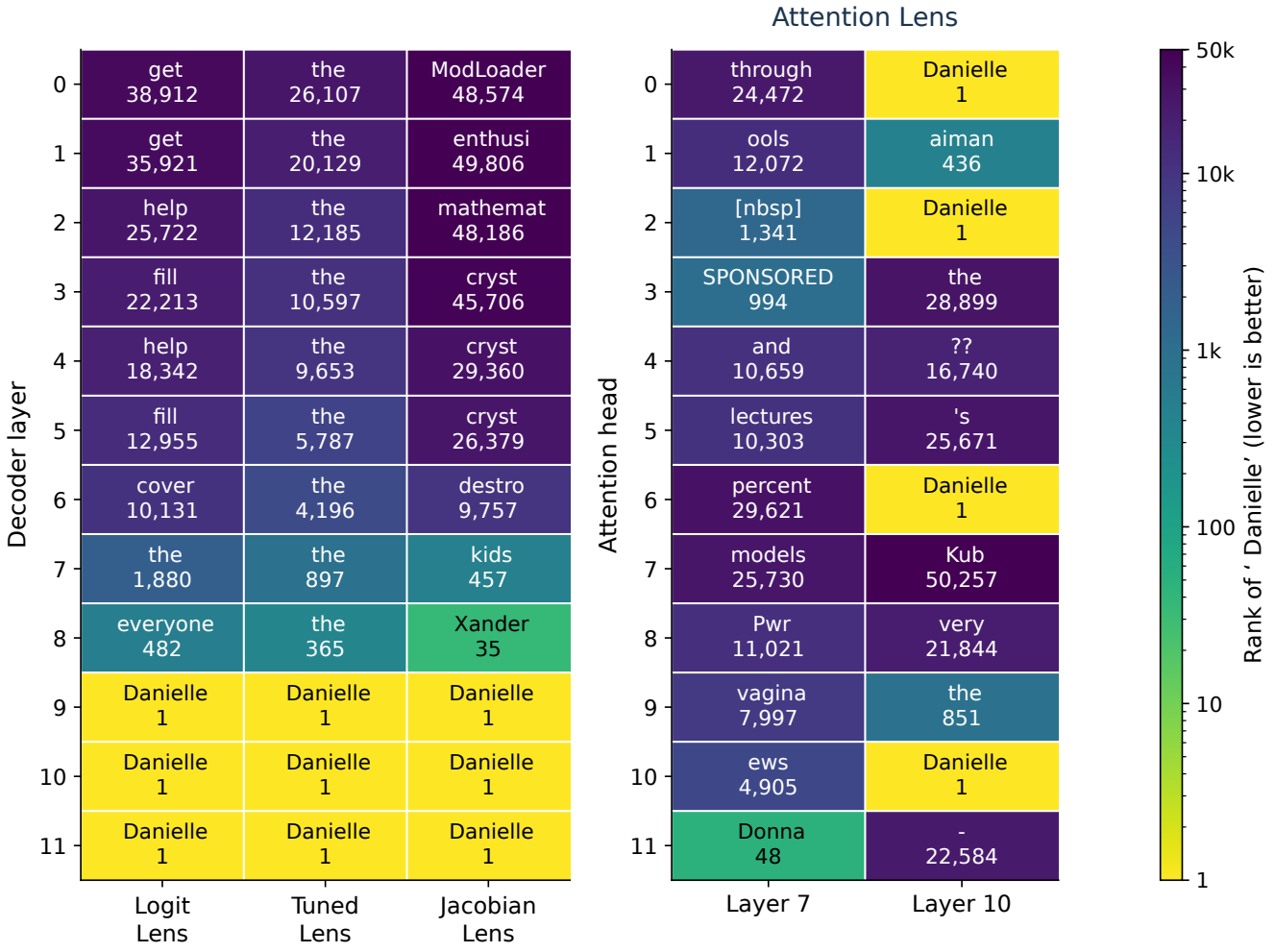


Figure 3. Residual and head readouts of the same IOI decision in GPT-2 small. Left: Logit, Tuned, and Jacobian Lens track Danielle at the final prediction position. Right: Attention Lens distinguishes heads within layers 7 and 10. Top-token labels and target ranks reveal agreement at late residual sites alongside substantial head-level variation.

define one shared fidelity metric. We therefore compare their visible readouts on matched inputs, rather than treating small-case distribution divergences as a method ranking. The separately fitted VLM translators and their independent validation settings are retained in Appendix A.

3.2 Lexical neighbors reveal associations rather than object labels

The implemented EmbedLens readout examines projected visual-token embeddings before the decoder (Fan et al. 2026). For visual token v_i and vocabulary embedding e_w, it reports neighbors ranked by $\cos(v_i, e_w) = v_i^\top e_w / (\|v_i\|\|e_w\|)$, with token norms available in the saved outputs. Figure 4 applies this readout to both GQA shirt images. The image annotations attach decoded lexical fragments to the selected visual patches.

A black-shirt patch associates with "black," while a yellow-shirt patch instead associates with "Blue." Successful and ambiguous neighbors therefore coexist under the same readout. This tool exposes associations cheaply and spatially, but neither cosine similarity nor a large norm is a calibrated object prediction. The

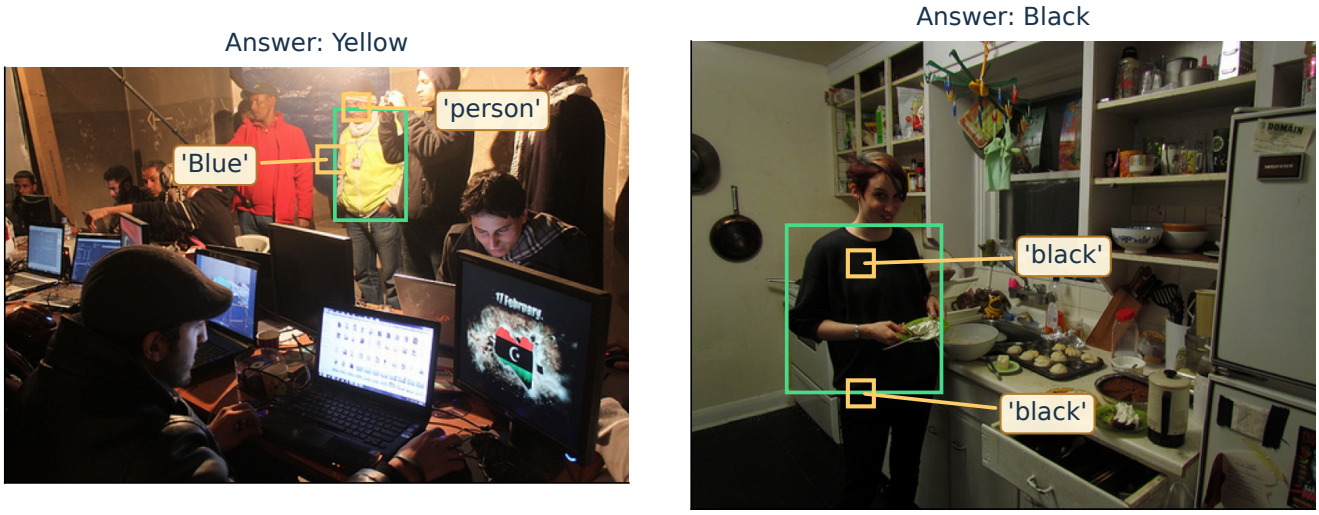


Figure 4. EmbedLens attaches lexical readouts to visual patches in LLaVA-1.5-7B. Amber patch boxes and leader lines connect the nearest vocabulary fragments to their projected visual tokens; green boxes mark the annotated shirt. These are embedding-space neighbors rather than independently detected object labels. The black-shirt example associates with "black," while yellow-shirt patches also yield unrelated fragments such as "Blue."

library currently implements this semantic readout; the paper's complete sink/dead/alive classification and pruning analysis are outside the demonstrated implementation.

3.3 Transfer readouts depend on the decoding prompt

Patchscopes moves a source representation into a specified target computation to make its content readable (Ghandeharioun et al. 2024). We use an identity mapping in LLaVA: an answer-position residual from a GQA source replaces a placeholder residual in a color-copy prompt. Figure 5 shows the source images and word-likelihood changes under a fixed target decoding context, relative to a zero-vector injection at that same target site.

The yellow source at layer 15 increases the complete-word Yellow log probability by 0.419 relative to zero injection, yet its top prediction is "is." Both sources produce "is" at layers 7, 15, and 23, and "->" at layer 31. This configuration fails to verbalize the color even though the source computation answers correctly. Unlike direct projection, a transfer readout can use a new decoding context, but its source site, mapping, and target prompt need validation. Likelihood changes alone do not establish successful information transfer.

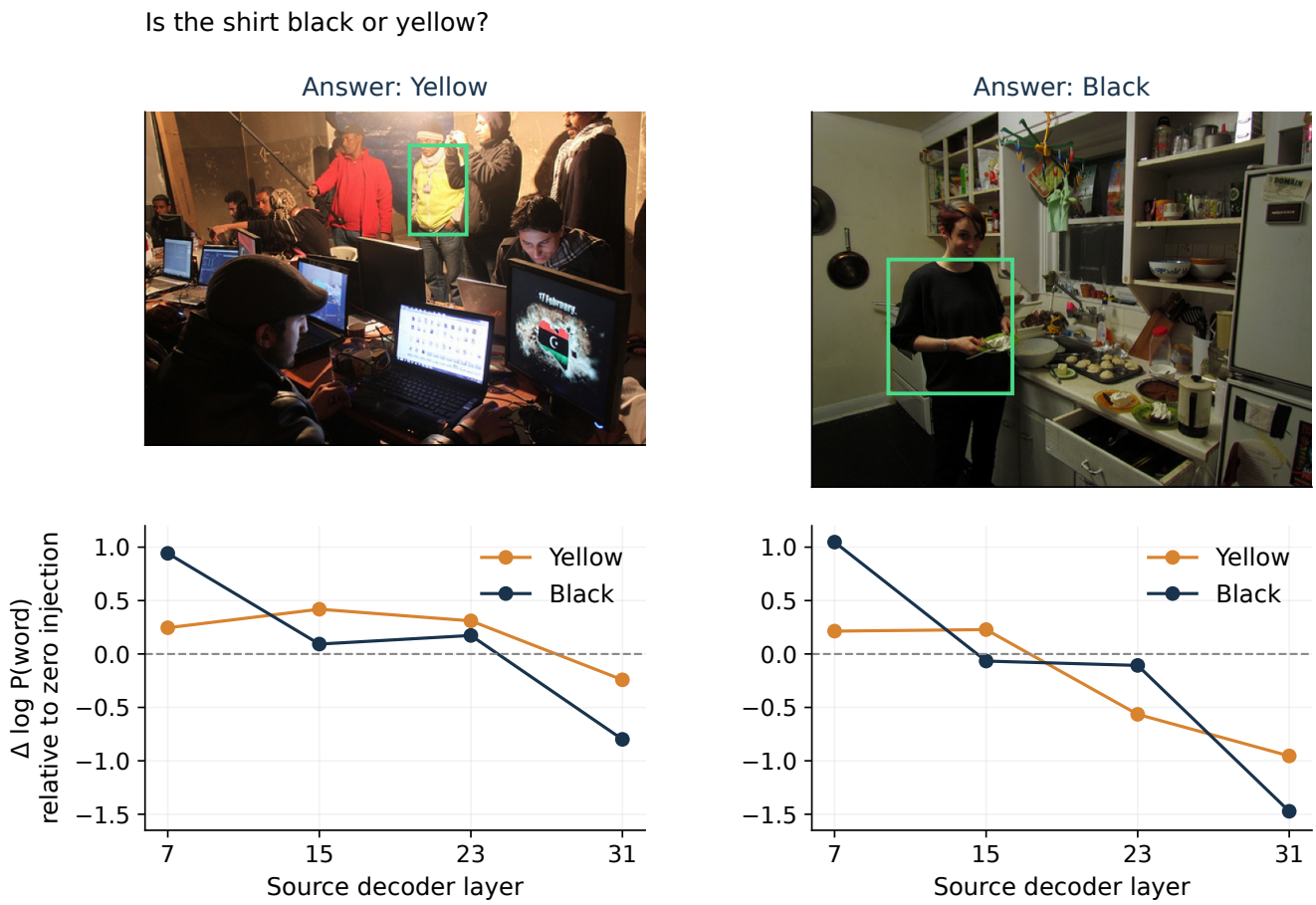


Figure 5. Patchscope reads a source residual through a new text context. In LLaVA-1.5-7B, the source answer-position residual replaces the placeholder after target layer 0 in a four-line color-copy prompt, `red -> red`, `blue -> blue`, `green -> green`, then `x ->`. Curves show changes in complete-word log probability relative to a same-site zero-vector injection; positive values indicate greater support for the plotted color. This configuration changes word likelihoods but fails to recover the shirt color as its top continuation.

4 Attention explanations

4.1 Propagation rules change the map for a fixed answer

Figure 6 compares raw attention and seven explanation methods on the same yellow-shirt GQA input across five VLMs. Reading across a row isolates the method; reading down a column isolates the model. Target-conditioned maps explain Yellow, including Qwen3-VL, whose native answer is Black.

The seven methods use different attribution and propagation rules: Rollout (Abnar and Zuidema 2020), attention Grad-CAM (Selvaraju et al. 2017), ATTATTR (Hao et al. 2021), TAM (Yuan et al. 2021), Beyond Intuition-H/C (Chen et al. 2023), and generic relevance (Chefer et al. 2021). Raw attention provides an additional reference. Rollout follows residual-augmented attention connections; Grad-CAM weights attention with a local target gradient; ATTATTR integrates sensitivity along an attention-probability path. The remaining methods use different gradient, value, or relevance-propagation rules. A common output shape makes their maps comparable visually, but does not make their scores equivalent.

The spatial differences are substantial even when the input and target remain fixed. Generic relevance concentrates more strongly within the annotated shirt region than rollout in each displayed model, whereas rollout spreads much of its map outside that object. A sharper map still does not establish a more faithful explanation: the box fractions describe this one input and each map has its own scale.

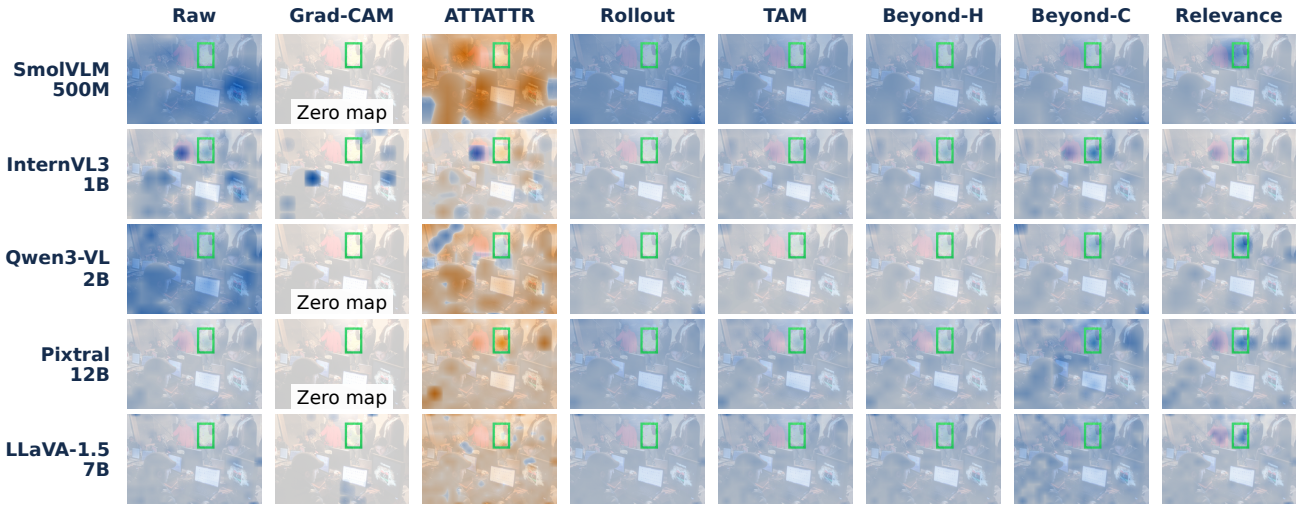


Figure 6. Attention explanations assign different spatial emphasis to the same yellow-shirt input. Rows compare five VLMs; columns compare raw attention and seven explanation methods. Target-conditioned methods explain Yellow. Blue indicates positive values and orange negative ATTATTR values; green boxes mark the shirt. Each map is independently normalized, so intensity compares positions within a map rather than score magnitude across maps. Original photographs are displayed at 20% opacity over opaque heatmaps. Raw attention, Grad-CAM, and ATTATTR inspect the last decoder attention layer; propagation methods span the decoder attention stack. "Zero map" denotes a measured null attribution.

ATTATTR adds a different distinction. Its signed results are predominantly negative in four models and more mixed in InternVL; discarding the sign would conceal that difference from positive-only maps.

4.2 A zero local map can coexist with visual dependence

SmolVLM, Qwen3-VL, and Pixtral have zero last-layer Grad-CAM maps, yet their TAM, Beyond Intuition, and generic relevance maps are nonzero. These methods inspect different paths or aggregate connections under different clipping rules. The actual residual replacement in Section 5.1 also changes SmolVLM's answer score. A zero local, rectified heatmap therefore does not establish that the image is irrelevant to the model. Local Grad-CAM offers a focused and inexpensive view; propagation examines broader dependencies while introducing additional modeling assumptions.

The adapters now expose the attention probabilities actually multiplied by values in the supported eager decoders. ATTATTR therefore uses the same public call across Llama, Qwen2, Qwen3-VL, and Mistral, without an architecture- specific user hook. A measured zero remains a result rather than an unavailable method. Qwen3.5's separate hybrid-stack limitations are discussed in Appendix A.2.

4.3 Deletion tests reveal disagreements hidden by plausible heatmaps

To check these methods on their original vision structure, we use pretrained ViT-B/16 and an author-released cat–dog image. This control is a vision classifier, not a VLM column omitted from the preceding comparison. Figure 7 combines the seven maps and their matched deletion curves in one figure. The image is unchanged between rows; the explained class changes from cat to dog. Target-independent rollout consequently stays the same, whereas target-conditioned tools can separate the objects.

Deleting 39 of 196 patches leaves cat probability at 65.1% under rollout, compared with 0.27% under Beyond-C; the original value is 68.6%. Conversely, the dog Grad-CAM map looks recognizable, but its deletion raises dog probability from 2.99% to 29.3%. A visual explanation and the effect of removing its selected regions can thus disagree. Deletion supplies an intervention-based diagnostic for choosing among maps, while remaining sensitive to the replacement baseline and removal of competing objects. The library supports both parts of this comparison rather than treating visual sharpness as sufficient evidence of faithfulness.

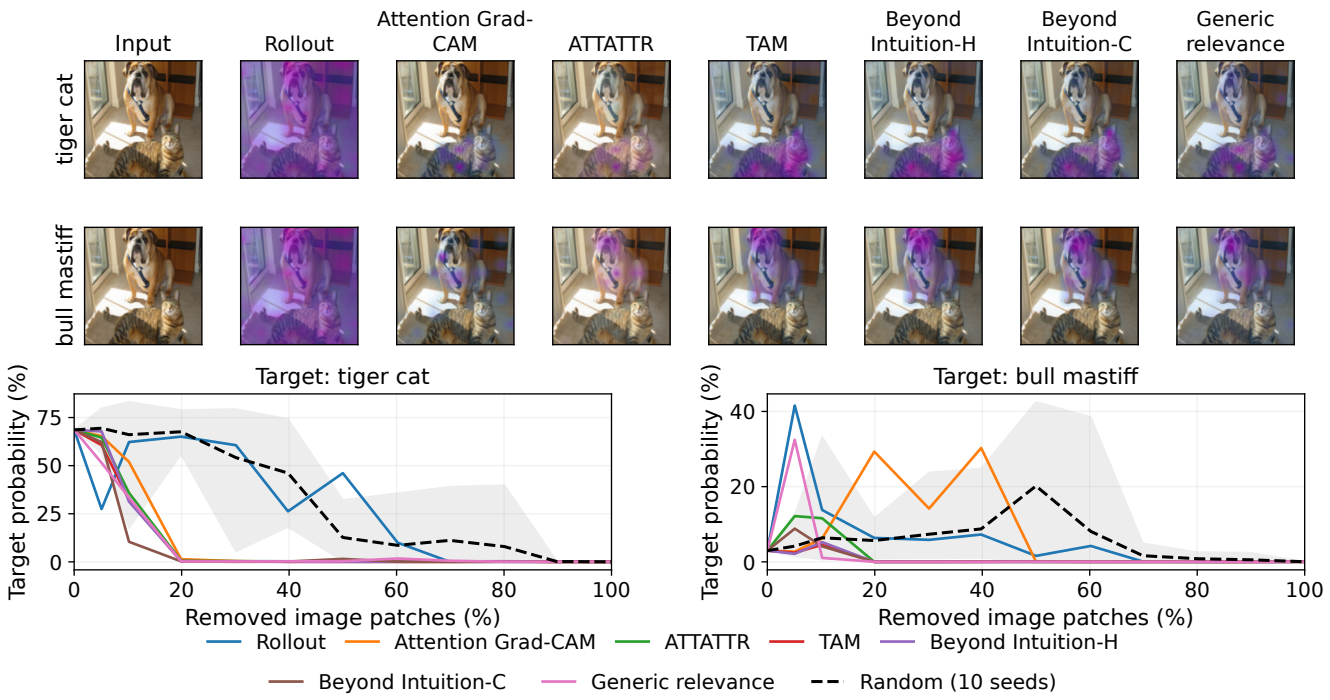


Figure 7. Object-conditioned attribution and deletion on ViT-B/16. Seven maps explain cat and dog in the same author-provided image. Curves measure class probability after replacing top-ranked patches with training-mean RGB. Rollout combines target-independent connections, while other tools can separate the classes; deletion exposes differences that map appearance alone does not resolve.

5 Causal interventions

5.1 Attribution patching can miss large replacement effects

Activation patching replaces a selected activation with its value from a reference input and recomputes downstream behavior (Meng et al. 2022). Attribution patching estimates the same replacement from a local gradient (Nanda 2023). For a residual $h_{\ell,S}$ at token set S and aligned reference value $d_{\ell,S}$,

$$\Delta_{\ell,S} = S(x; h_{\ell,S} \leftarrow d_{\ell,S}) - S(x), \qquad \widehat{\Delta}_{\ell,S} = \left\langle \nabla_{h_{\ell,S}} S(x),\ d_{\ell,S} - h_{\ell,S} \right\rangle .$$

We compare these effects on the same GQA input in SmolVLM and LLaVA. The reference input is a gray image with identical dimensions, prompt, and answer prefixes. Figure 8 distinguishes replacement of all visual states from replacement of the answer-predicting states. These are hidden-state interventions: the image and text of the patched run stay fixed. Answer-predicting states are the residual vectors used to predict each piece of the scored answer; a multi-piece answer can involve more than one position. Actual effects are measured at seven layers; first-order estimates cover every layer. The diagram shows the two alternatives, tested in separate runs.

SmolVLM's layer-5 visual replacement changes log $P(\text{Yellow})$ by -0.498, compared with an estimate of -0.067. LLaVA's corresponding effect is -1.646, while the estimate is $+0.100$, reversing the sign. Large blank-image replacements leave the neighborhood described by a local gradient. Attribution patching can cheaply nominate candidate sites, but these results motivate testing important candidates with actual replacement. They do not imply that every smaller head- or token-level estimate will fail.

5.2 The intervention unit determines the apparent locus of dependence

Image-slot replacement is strongest early in the yellow-shirt case; replacing prediction slots has larger late effects. The final visual-slot effect is zero because no remaining decoder layer transports the change into

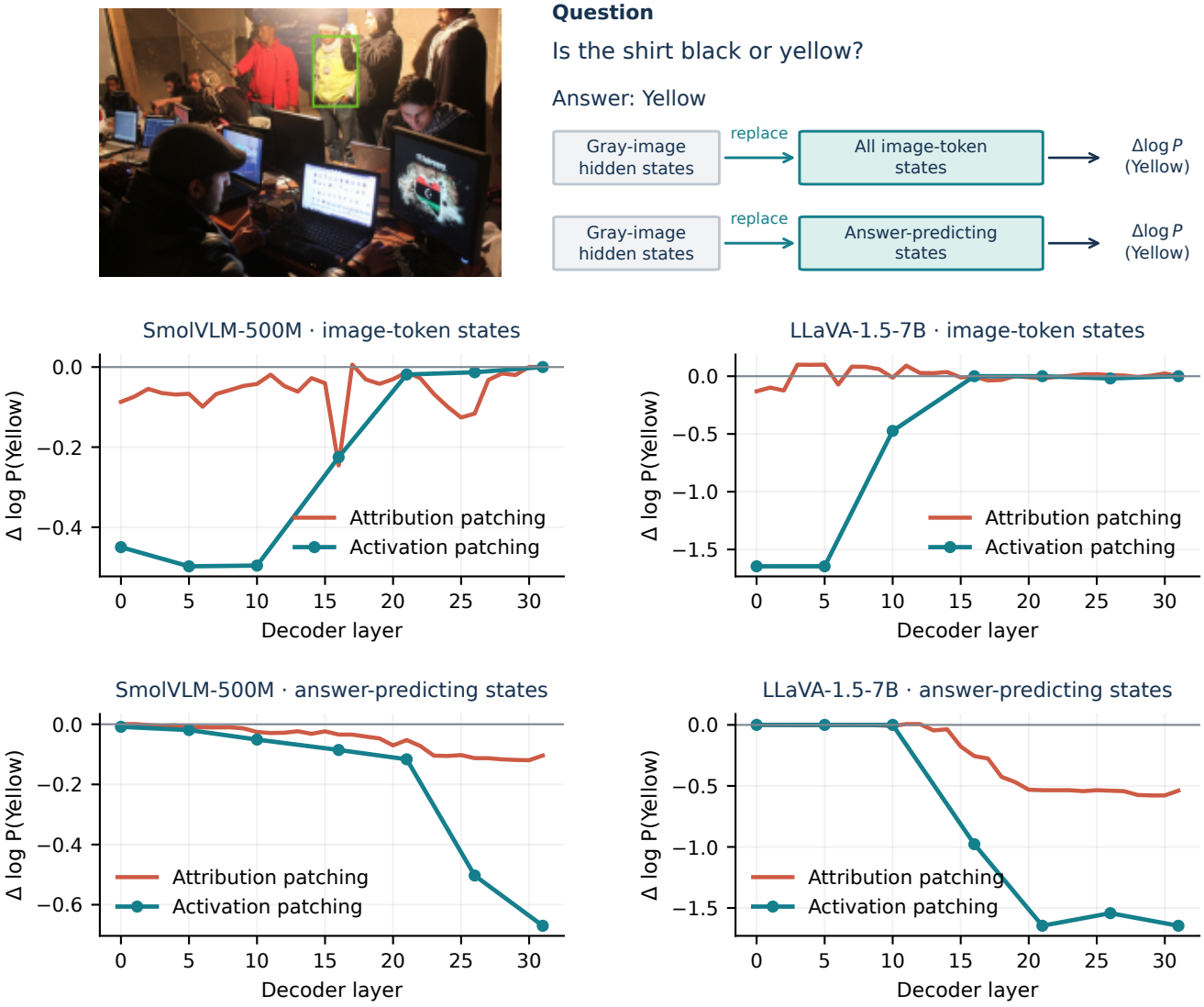


Figure 8. Attribution patching can miss effects measured by activation patching. Gray-image hidden states replace selected states in the real-image run at one decoder layer. The schematic rows are separate interventions: all image-token states or the states that predict the answer's token pieces. Teal markers measure changes in the complete-answer log probability; red curves estimate the same changes from gradients. Negative values reduce support for Yellow. The estimates miss large early image-state and late answer-state effects.

an answer position. That zero is a consequence of the intervention location, rather than evidence that the model ignored the image.

Figure 9 refines the experiment to one native text position at a time. Rows are layers and columns are question or assistant-boundary subwords; actual and estimated effects share a signed scale. Replacing the question word "yellow" at an intermediate layer changes the answer score in both models. The largest late effects occur at the final prediction position, whose role makes this sensitivity expected. This distinguishes sensitivity at question-token sites from disruption of the answer readout itself.

The black-shirt input provides a useful contrast (Figures 16 and 17). LLaVA's largest prediction-slot effect is only -0.172, compared with -1.646 on Yellow. The maps therefore characterize a specified input and corruption. Neither two examples nor a visually similar layer pattern establishes a universal pathway or model-level robustness ranking.

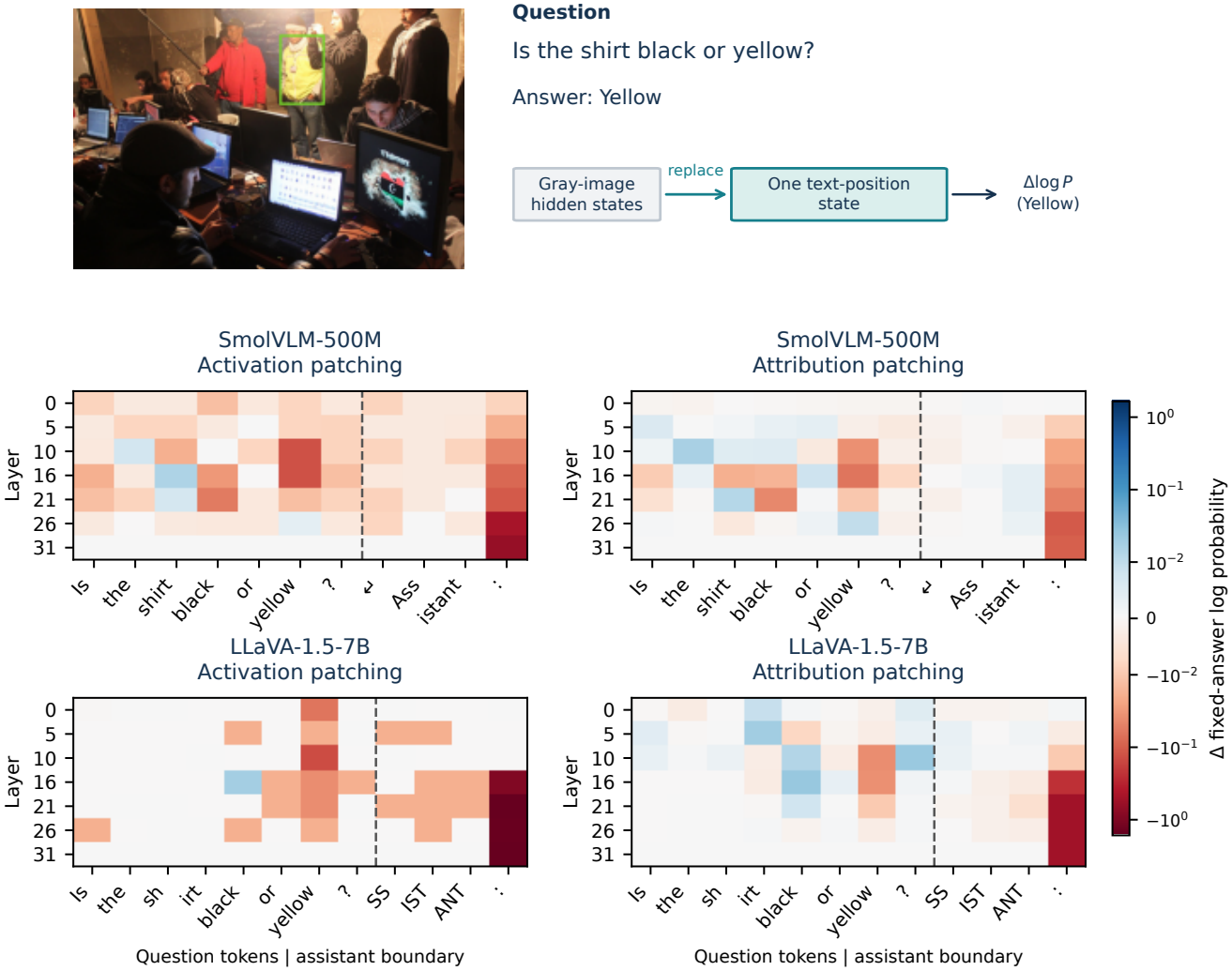


Figure 9. Single-position replacement separates question-word effects from prediction-site sensitivity. Each cell replaces one native text-position state with its corresponding gray-image state at the indicated layer, as depicted in the schematic. Columns cover question and assistant-boundary subwords. Red reduces the complete-answer log probability of Yellow; blue increases it. Activation patching measures the replacement effect and attribution patching estimates it, using the same signed scale. Contrast with the joint replacements in Figure 8.

5.3 Knockout tests information routes, while path patching isolates senders

Attention knockout blocks selected key-to-query connections before softmax, allowing downstream computation to recompute (Zhang et al. 2025). On the yellow-shirt GQA input, we compare image-to-question, image-to-prediction, and question-to-prediction routes over windows of up to nine layers. Figure 10 plots the fixed first-answer-token probability and compares target-region and other-image keys. The displayed question identifies the input. This paper-aligned demonstration uses a single-word-or-phrase instruction and a first-subword metric, rather than the shared complete-answer score; Appendix A records the full instruction.

The image-to-question route is more disruptive than the direct image-to-prediction route on this example. Other-region blockade can instead increase answer support. Knockout tests dependence on a route without requiring a reference input, but changes attention normalization and depends on the mask/window. The target region contains 20 keys versus 556 outside it, so a size-matched control would be required to attribute the difference specifically to grounding.

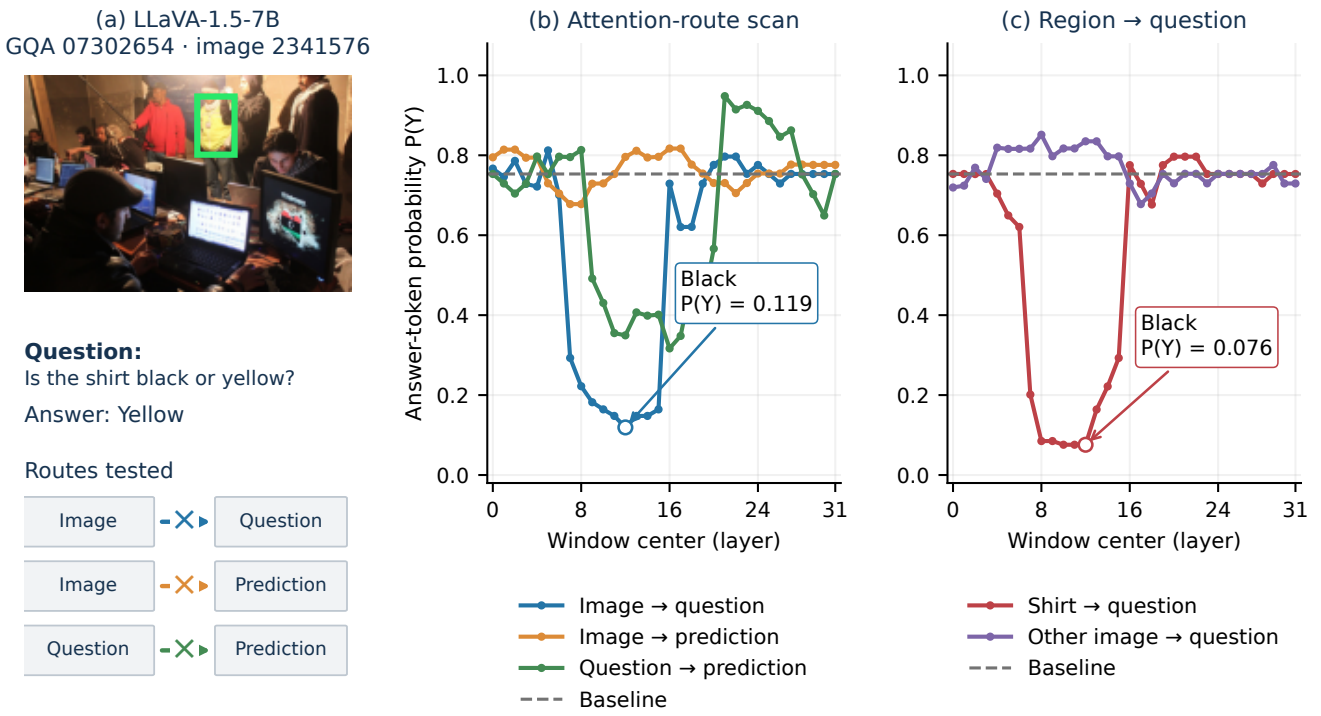


Figure 10. Attention Knockout reveals route-specific dependence in LLaVA-1.5-7B. Blocking image-to-question connections at layers 8–16 lowers the fixed Yellow first-subword probability from 0.753 to 0.119 and makes Black the next-token prediction. Blocking shirt keys lowers it to 0.076; region sizes differ, so these curves describe the selected masks rather than controlled localization accuracy.

IOI-style path patching asks a narrower question (Wang et al. 2023). A controlled base run replaces a sender head with its reference-input output while freezing other attention-head outputs to base values. MLPs and normalization recompute; a fresh base run then replays only the resulting receiver activation. Figure 11 sweeps sender heads on a paired MMVP wing-status input in five VLMs, using the final prediction-position residual as receiver.

The largest negative effects include Qwen3-VL L23.H6 (-0.054), Pixtral L34.H12 (-0.080), and LLaVA L14.H3 (-0.064). The heatmaps identify influential senders under this receiver and freezing rule. They are not an automatically discovered complete circuit or functional labels such as "name mover." Path patching offers tighter control of selected paths than unrestricted residual replacement, at the cost of more forwards and a receiver-dependent question.

5.4 Circuit attribution requires measured recovery

EAP and EAP-IG rank independently replaceable connections in an explicit computation graph; EAP-IG integrates gradients along the input-embedding path (Hanna et al. 2024). We retain the paper-aligned GPT-2 IOI graph, rather than substituting VLM residual nodes for unimplemented edges. Figure 12 evaluates both rankings by actual edge-retention interventions on the same 32,491-edge graph. Excluded edges receive reference messages, while retained edges remain live. Recovery is measured as

$$R(E) = \frac{m(E) - m(\text{reference})}{m(\text{clean}) - m(\text{reference})},$$

where m is the Danielle-minus-Andrew logit margin. This quantity can fall below zero or exceed one; it is not a probability or a sum of attribution scores.

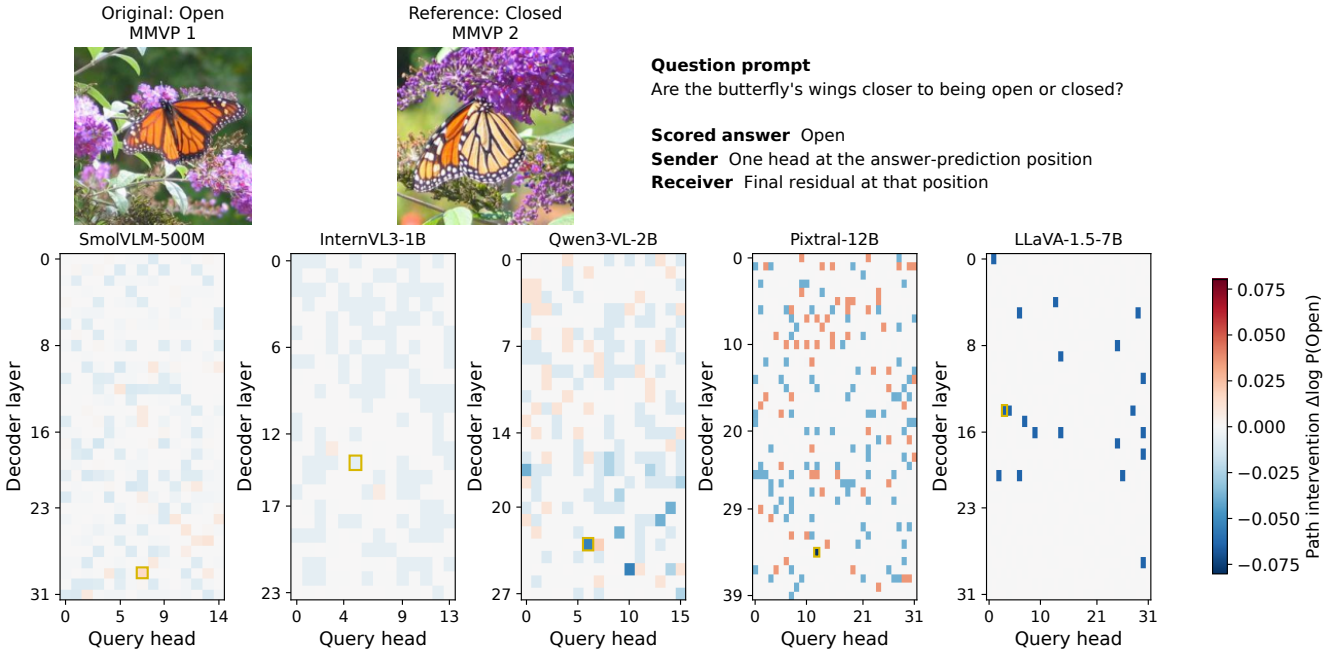


Figure 11. Controlled sender-head paths on a paired butterfly-wing input. Each cell replaces one sender using its reference-image value at the prediction position, with other head outputs frozen and only the recomputed final residual replayed. Other head outputs are frozen during the controlled run; MLP and normalization recompute, and only the resulting receiver is injected in a fresh run. Signed effects compare the same configured receiver across five VLMs. Yellow outlines identify the largest absolute effect within each model; blue decreases and red increases support for Open.

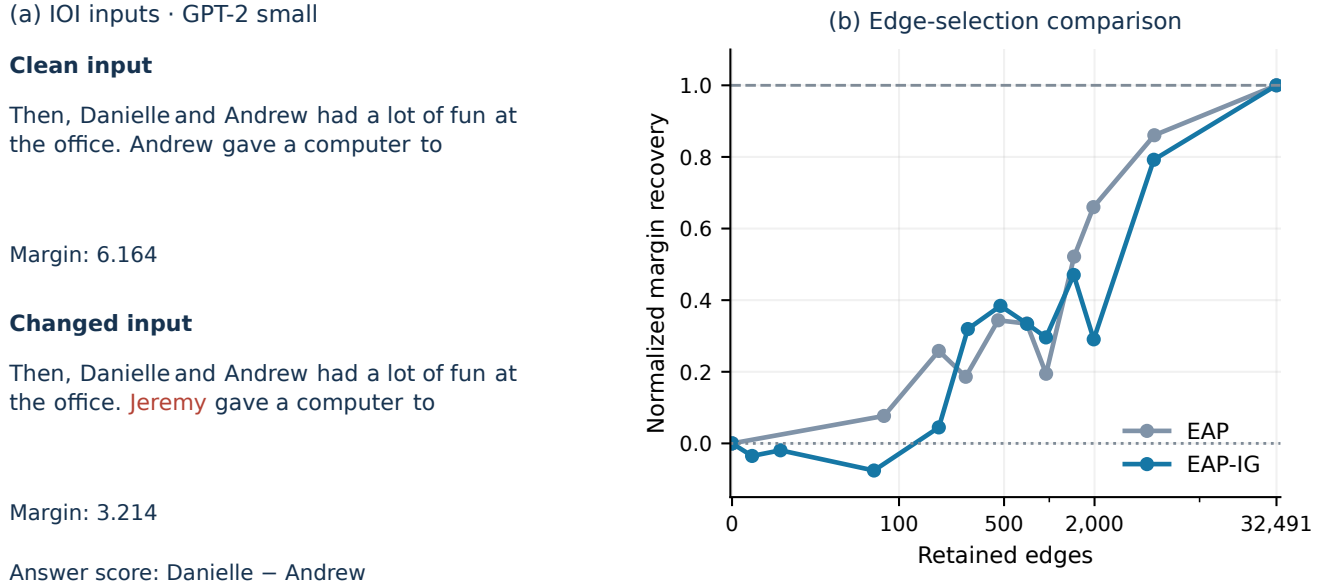


Figure 12. Attribution rankings differ in actual IOI circuit recovery. EAP and input-path EAP-IG select connections in the same GPT-2 graph; each curve point is a fresh retained-edge intervention. Both complete graphs recover the clean margin, but EAP-IG does not uniformly outperform EAP on this single prompt pair.

At 184 retained edges, EAP recovers 0.258 of the original-reference margin difference, while EAP-IG recovers 0.045. Integrating gradients therefore does not guarantee a better selected circuit on each instance. The practical benefit of this comparison is the separation of ranking and validation: a high attribution score proposes a connection, and edge retention tests the proposed subgraph. The graph-specific intervention contract and single-pair evaluation limit what can be inferred about general circuit quality.

5.5 Steering depends on both model and strength

VSV, the visual steering component of VISTA, adds an image-derived residual direction during generation (Li et al. 2025). Figure 13 holds a COCO image and its description request fixed across LLaVA, SmolVLM and Qwen3-VL. Each row compares the model's baseline with two smaller steering scales. Exact excerpts retain the changed content, and ellipses indicate omissions. These outputs examine generation changes directly, rather than necessity under a corruption.

COCO2014 · image 310196



Prompt: Please help me describe the image in detail.

	Baseline	VSV · lower λ	VSV · higher λ
LLaVA-1.5 7B	**λ = 0** … The snowy terrain is dotted with various tracks, indicating that the slope is frequently used by snowboarders and skiers. …	**λ = 0.02** … The snowy landscape is dotted with several snow-covered rocks scattered throughout the scene, adding to the wintry atmosphere. …	**λ = 0.05** … The snow-covered slope is a popular skiing and snowboarding spot, with a few other people scattered around the area, likely enjoying the winter sports. …
SmolVLM 500M	**λ = 0** … The snowboard is a standard snowboard, characterized by its flat, rectangular shape and a single, large, flat, circular nose. …	**λ = 0.02** … The individual is captured mid-air, … They are wearing a red and black outfit, … The person is holding the snowboard with both hands, …	**λ = 0.05** … The individual is captured mid-air, … They are wearing a red and black outfit, … performing a snowboard trick. …
Qwen3-VL 2B	**λ = 0** … The central focus is a single skier, seen from above, who is carving down the slope. …	**λ = 0.02** … The skier, positioned in the lower-right quadrant of the image, is the main focal point. …	**λ = 0.05** … A single person is the focal point of the image, positioned in the lower-right quadrant, actively skiing down the slope. …

Figure 13. The same image-description request under residual VSV in three models. Rows hold the model fixed; columns compare its baseline with two steering strengths, printed in each cell. Excerpts preserve the generated wording, with ellipses marking omitted text. Green highlights newly mentioned details supported by the image; red highlights selected claims without image support. Uncolored text is not fully audited. The common intervention follows the raw residual rule, without SLA.

LLaVA illustrates why fluency and grounding need separate inspection. Its baseline already mentions tracks but infers frequent use of the slope. At strength 0.02, it instead mentions visible rocks; at 0.05, it invents other people. All three descriptions are grammatical. The retained 0.17 stress run repeats snow-related text even after exact text-token alignment.

SmolVLM changes from a detailed equipment description to a maneuver description, adding the visible red-and-black clothing but also an unsupported airborne pose and board-holding claim. Qwen3-VL retains a single-rider account while changing its spatial framing and track description. The same raw scale therefore changes different claims in different models; added visual detail and newly invented content can occur within the same description.

Our intervention uses the paper's raw block-residual rule, without SLA. The relative additive displacement depends on $\lambda\|v_\ell\|/\|h_\ell\|$, so identical scales need not produce equivalent perturbations across models. The released VISTA code instead uses normalized MLP-output directions and additional direction processing. Its numerical scale cannot be assumed to transfer to this implementation.

Figure 14 tests local evidence under fixed prefixes rather than comparing the changing prefixes of those descriptions. The maps give person/car token ranks at the same 80 baseline prefixes; the bottom curve holds one prefix and prediction position fixed across strengths.

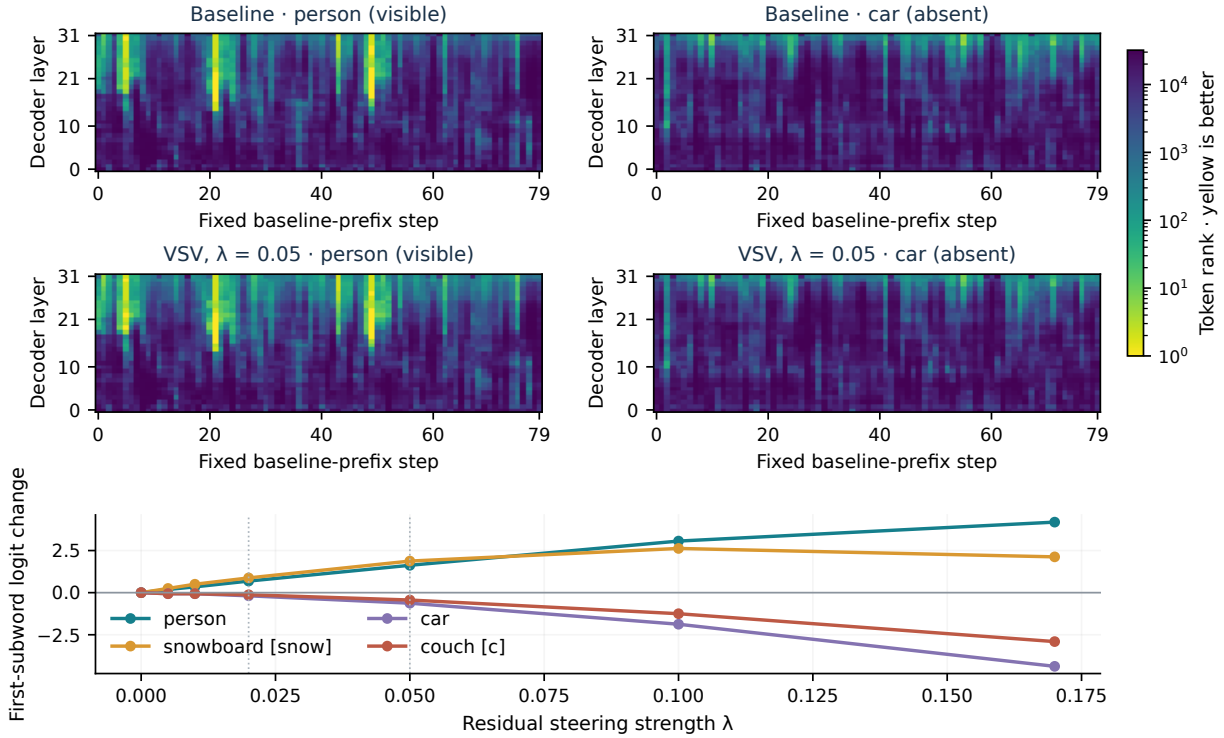


Figure 14. Object-token support under fixed prefixes does not guarantee a grounded description. In the matched LLaVA run, baseline and steered person/car heatmaps share the same 80 baseline prefixes and logarithmic rank scale; brighter colors indicate better ranks. The strength curves fix the prefix "The image captures a snowboarder skillfully riding" and measure first-subword logit changes, not complete-word probabilities. At the highest retained strength, favorable person/car responses coexist with repetitive free generation.

At strength 0.17, person's logit increases by 4.19 and car's decreases by 4.38, despite the repetitive free generation. A favorable object-token response therefore need not yield a grounded, fluent description. VSV offers a direct way to alter visual influence, but its strength requires evaluation against both grounding and decoding quality. This is a Steering/VSV study; the SLA component is absent, so it does not represent complete VISTA reproduction.

Across these intervention groups, the library keeps the change and its question explicit. Residual replacement tests a counterfactual activation; knockout tests a blocked route; path patching controls sender influence; edge retention evaluates a proposed circuit; steering changes generation. Matched comparisons clarify differences within those questions, while separate studies preserve the meaning of methods with incompatible interventions.

References

Abnar, Samira and Willem Zuidema (2020). "Quantifying Attention Flow in Transformers." In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pp. 4190–4197. DOI: 10.18653/v1/2020.acl-main.385.

Belrose, Nora, Igor Ostrovsky, Lev McKinney, Zach Furman, Logan Smith, Danny Halawi, Stella Biderman, and Jacob Steinhardt (2023). "Eliciting Latent Predictions from Transformers with the Tuned Lens." In: *arXiv preprint*. arXiv: 2303.08112.

Chefer, Hila, Shir Gur, and Lior Wolf (2021). "Generic Attention-model Explainability for Interpreting Bi-Modal and Encoder-Decoder Transformers." In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 397–406.

Chen, Jiamin, Xuhong Li, Lei Yu, Dejing Dou, and Haoyi Xiong (Feb. 2023). "Beyond Intuition: Rethinking Token Attributions Inside Transformers." In: *Transactions on Machine Learning Research*.

Fan, Yingqi, Junlong Tong, Anhao Zhao, and Xiaoyu Shen (2026). "What Do Visual Tokens Really Encode? Uncovering Sparsity and Redundancy in Multimodal Large Language Models." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11987–11997.

Ghandeharioun, Asma, Avi Caciularu, Adam Pearce, Lucas Dixon, and Mor Geva (2024). "Patchscopes: A Unifying Framework for Inspecting Hidden Representations of Language Models." In: *Proceedings of the 41st International Conference on Machine Learning*. Vol. 235. Proceedings of Machine Learning Research, pp. 15466–15490.

Gurnee, Wes, Nicholas Sofroniew, Adam Pearce, Mateusz Piotrowski, Isaac Kauvar, Runjin Chen, Anna Soligo, Paul Bogdan, Euan Ong, Rowan Wang, T. Ben Thompson, David Abrahams, Subhash Kantamneni, Emmanuel Ameisen, Joshua Batson, and Jack Lindsey (July 6, 2026). *Verbalizable Representations Form a Global Workspace in Language Models*. Transformer Circuits. URL: https://transformer-circuits.pub/2026/workspace/index.html.

Hanna, Michael, Sandro Pezzelle, and Yonatan Belinkov (2024). "Have Faith in Faithfulness: Going Beyond Circuit Overlap When Finding Model Mechanisms." In: *Conference on Language Modeling*. arXiv: 2403.17806.

Hao, Yaru, Li Dong, Furu Wei, and Ke Xu (2021). "Self-Attention Attribution: Interpreting Information Interactions Inside Transformer." In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 35. 14, pp. 12963–12971. DOI: 10.1609/aaai.v35i14.17533.

Li, Zhuowei, Haizhou Shi, Yunhe Gao, Di Liu, Zhenting Wang, Yuxiao Chen, Ting Liu, Long Zhao, Hao Wang, and Dimitris N. Metaxas (2025). "The Hidden Life of Tokens: Reducing Hallucination of Large Vision-Language Models via Visual Information Steering." In: *Proceedings of the 42nd International Conference on Machine Learning*. Vol. 267. Proceedings of Machine Learning Research.

Meng, Kevin, David Bau, Alex Andonian, and Yonatan Belinkov (2022). "Locating and Editing Factual Associations in GPT." In: *Advances in Neural Information Processing Systems*. Vol. 35.

Nanda, Neel (2023). *Attribution Patching: Activation Patching at Industrial Scale*. URL: https://www.neelnanda.io/mechanistic-interpretability/attribution-patching.

nostalgebraist (2020). *Interpreting GPT: The Logit Lens*. URL: https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens.

Sakarvadia, Mansi, Arham Khan, Aswathy Ajith, Daniel Grzenda, Nathaniel Hudson, André Bauer, Kyle Chard, and Ian Foster (2023). "Attention Lens: A Tool for Mechanistically Interpreting the Attention Head Information Retrieval Mechanism." In: *Proceedings of the Sixth BlackboxNLP Workshop*. arXiv: 2310.16270.

Selvaraju, Ramprasaath R., Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra (2017). "Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization." In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 618–626.

Wang, Kevin, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt (2023). "Interpretability in the Wild: A Circuit for Indirect Object Identification in GPT-2 Small." In: *International Conference on Learning Representations*.

Yuan, Tingyi, Xuhong Li, Haoyi Xiong, Hui Cao, and Dejing Dou (2021). "Explaining Information Flow Inside Vision Transformers Using Markov Chain." In: *First Workshop on eXplainable AI Approaches for Debugging and Diagnosis, NeurIPS*.

Zhang, Zhi, Srishti Yadav, Fengze Han, and Ekaterina Shutova (2025). "Cross-modal Information Flow in Multimodal Large Language Models." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. arXiv: 2411.18620.

Data resources

Hudson, Drew A. and Christopher D. Manning (2019). "GQA: A New Dataset for Real-World Visual Reasoning and Compositional Question Answering." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6700–6709.

Kamoi, Ryo, Yusen Zhang, Sarkar Snigdha Sarathi Das, Ranran Haoran Zhang, and Rui Zhang (2025). "VisOnlyQA: Large Vision Language Models Still Struggle with Visual Perception of Geometric Information." In: *Conference on Language Modeling*. arXiv: 2412.00947.

Liu, Fangyu, Guy Emerson, and Nigel Collier (2023). "Visual Spatial Reasoning." In: *Transactions of the Association for Computational Linguistics* 11, pp. 635–651. DOI: 10.1162/tacl_a_00566.

Merity, Stephen, Caiming Xiong, James Bradbury, and Richard Socher (2017). "Pointer Sentinel Mixture Models." In: *International Conference on Learning Representations*.

Tong, Shengbang, Zhuang Liu, Yuexiang Zhai, Yi Ma, Yann LeCun, and Saining Xie (2024). "Eyes Wide Shut? Exploring the Visual Shortcomings of Multimodal LLMs." In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*.

Zhu, Yukun, Ryan Kiros, Rich Zemel, Ruslan Salakhutdinov, Raquel Urtasun, Antonio Torralba, and Sanja Fidler (2015). "Aligning Books and Movies: Towards Story-Like Visual Explanations by Watching Movies and Reading Books." In: *Proceedings of the IEEE International Conference on Computer Vision*, pp. 19–27.

A Experimental details

A.1 Checkpoints and input contracts

The shared VLM runs use Torch 2.6 and Transformers 5.3, BF16 weights, eager attention, and full-sequence uncached forwards. Pixtral uses a 256-pixel longest edge and Transformers' native Mistral regex correction, applied after tokenizer loading to avoid a duplicate-keyword bug in version 5.3. Pinned files are unchanged. Checkpoint identities are listed in Table 4; full revisions bind the model artifacts used by every archived run. Weights and bulk datasets are stored outside the repository under `/bigdata/gexin/vlm_probing`.

Table 4. Pinned pretrained checkpoints for the shared VLM experiments.

Checkpoint	Revision
`HuggingFaceTB/SmolVLM-500M-Instruct`	`a7da5b986cb59b408707209984f360a5f4ad7e47`
`OpenGVLab/InternVL3-1B-hf`	`014c0583a0d4bedf29fbe2dbff4f865eb998e171`
`Qwen/Qwen3-VL-2B-Instruct`	`89644892e4d85e24eaac8bacfd4f463576704203`
`mistral-community/pixtral-12b`	`c2756cbbb9422eba9f6c5c439a214b0392dfc998`
`llava-hf/llava-1.5-7b-hf`	`b234b804b114d9e37bb655e11cbbb5f5e971b7a9`
`Qwen/Qwen3.5-4B` (appendix)	`851bf6e806efd8d0a36b00ddf55e13ccb7b8cd0a`

GQA photographs come from the official image archive; questions and boxes come from the cross-modal information-flow authors' choose-attribute subset. The image IDs are 2341576 and 2342134, with boxes $(264, 33, 57, 89)$ and $(76, 147, 124, 134)$ in (x, y, w, h) coordinates. MMVP uses the first released image pair. Its HF split is named `train`, but the images are released evaluation cases, not lens-fitting examples. VSR uses rows 0 and 1 of the zero-shot test split. The first row's car is partly visible as a dashboard, so it is unsuitable for a clear external-car localization claim. VisOnlyQA uses row 0 of the real geometry-triangle and chart-extraction splits, retaining its official no-reasoning prompts.

Shared-matrix GQA/MMVP/VSR prompts prescribe sorted answer options and no explanations. SmolVLM uses its native leading-space continuation; other word answers use their recorded bare-word encoding. LLaVA's Yellow has two answer pieces. The shared score includes all pieces. Original images, padded images, literal chat prompts, answer IDs, expanded positions, input hashes, and native generations are archived in `docs/report/v10/results`; previous records remain in `docs/report/results`.

A.2 Hybrid attention in Qwen3.5

The preceding experiment also evaluated Qwen3.5-4B (`Qwen/Qwen3.5-4B`) on the eight shared inputs; its pinned revision is included in Table 4. Its 32 decoder blocks contain 24 Gated DeltaNet blocks and eight full-attention blocks. Residual lenses, activation patching, attribution patching, and steering remain defined at all block outputs. Local attention methods can inspect the full-attention blocks when the required probability sites are exposed. Standard attention rollout, however, assumes an attention matrix at every propagation step. Skipping DeltaNet blocks would omit their mixing and does not reproduce the standard method. Extending rollout requires defining and validating propagation through those blocks. Likewise, IOI-style path patching requires a controlled output-freezing contract for every component in the stack, which the current DeltaNet adapter does not supply. These structural boundaries differ from a missing editable softmax tap in a conventional decoder. The main comparison uses Pixtral to retain full-stack methods across all columns; the Qwen3.5 residual results remain in the archived supplementary experiment in `docs/report/results/qwen35`.

A.3 Lens fitting and readout scope

The two VLM Tuned Lenses use 1,024 COCO2014 training images and 256 separate validation images, with eight caption prediction positions per image. The frozen model supplies 8,192 training and 2,048 validation positions. At layers 7, 15, and 23, FP32 affine translators minimize teacher-distribution KL with Adam, learning rate 3×10^{-4}, batch size 64, a maximum of 12 epochs, patience 3, improvement threshold 10^{-4}, and seed 20261007. Epoch zero is a valid model-selection candidate. Shared report cases are excluded from fitting; recorded SmolVLM overlap checks include image hashes and VSR COCO source IDs. LLaVA artifacts were fitted previously using SDPA and image-mean padding, then replayed under the report's eager/gray conditions.

Table 5 reports independent validation fidelity. It records calibration quality on held-out caption positions, rather than ground-truth answer accuracy or a comparison of different lens objectives.

Table 5. Tuned Lens validation KL in nats per selected position. Initial and selected translators are evaluated on the same held-out caption positions.

Model	Layer	Initial KL	Selected KL	Epoch
SmolVLM	7	14.3279	1.9050	11
SmolVLM	15	11.4524	1.4967	12
SmolVLM	23	6.0436	0.7889	11
LLaVA	7	7.8680	1.3609	12
LLaVA	15	5.9005	0.9137	12
LLaVA	23	4.5529	0.3782	11

The GPT-2 Tuned Lens is imported from the authors and mapped to post-block sites. Attention Lens is fitted on 4,096 BookCorpus contexts (Zhu et al. 2015), with 512 WikiText-2 validation contexts for cross-domain validation (Merity et al. 2017). BookCorpus uses the historical HF archive's `books_large_p2.txt` prefix, not BookCorpusOpen; each context has 96 tokens and supervises the final position. WikiText uses `Salesforce/wikitext`, `wikitext-2-raw-v1`; its validation and test splits supply 512 contexts each. Jacobian Lens calibrates all twelve GPT-2 residual layers on 256 WikiText-2 training contexts. The preceding experiment selected only layers 3, 7, and 10; its uncalibrated rows were a configuration choice, not a method limitation. The Jacobian readout estimates causal transport into the final residual basis; its scores are not calibrated next-token probabilities. The available artifacts support the text study, not a five-VLM calibrated Jacobian/Attention Lens comparison.

EmbedLens reports lexical neighbors of projected visual input embeddings, without category thresholds for the full paper's token classification. EmbedLens and Patchscope retain the earlier single-word-or-phrase GQA instruction. Images and questions are shared, but their source residuals are not pooled with the matrix's forced-option run. Patchscope uses source layers 7, 15, 23, and 31 at the answer-prediction position, replacing the target placeholder after layer 0 with an identity mapping. Its zero-vector control uses the same target site and scoring prompt. Both are readout studies, not completed pruning or cross-model transfer trials.

A.4 Attention implementations and spatial display

Single-layer Grad-CAM and ATTATTR use the last available native attention layer; decoder propagation methods start at layer 0 and require a consecutive stack. Qwen3-VL propagation does not include the vision encoder, projector, or DeepStack injections as separate graph edges. Qwen3.5 has eight full-attention layers at indices 3, 7, 11, 15, 19, 23, 27, and 31, interleaved with 24 DeltaNet layers. Skipping the latter is not a full-stack rollout.

TAM and Beyond input integrals use 20 right-Riemann samples at $\alpha = 1/20, \ldots, 1$. SmolVLM interprets exactly zero processed image pixels as a padding-image sentinel; excluding that endpoint avoids removing its image token sequence. This quadrature adaptation is applied consistently across the VLM comparison. ATTATTR retains the official-code left-sample convention with 20 samples. TAM uses residual normalization by $1/2$.

The raw native patch grid is interpolated in the recorded padded-image frame and then cropped using the original padding offsets. Each spatial map is normalized by its own native-grid maximum absolute value, without renormalization after cropping. In Figure 6, heatmaps remain fully opaque and the original photograph is overlaid at 20% opacity. Color intensity follows a monotone power transform with exponent 0.4 to make weak spatial variation visible. The palette runs from white at zero to blue for positive values and orange for negative ATTATTR values. This display transform does not change the archived scores; measured zero maps are explicitly labeled. Shirt-mass diagnostics integrate absolute values in the cropped display coordinates and are not benchmark localization scores. Original-grid numerical arrays remain available in the archives.

The native vision control uses the original timm/JAX checkpoint `jx_vit_base_p16_224-80ecf9dd.pth`, losslessly converted to a native HF 12-layer, 14×14-patch classifier. The Google ViT identifier supplies configuration and class labels, not the measured weights. ViT CAM/ATTATTR explain layer 11 at CLS; ATTATTR uses 20 right samples, TAM/Beyond use 20 endpoint-inclusive samples, and Beyond-H/C start at layers 3/5. Its cat–dog and elephant–zebra images are author-released illustrative assets, not a benchmark subset. Deletion sets selected processed patches to zero, equivalent to training-mean RGB; ten seeded random orders define the displayed envelope. Original ViT paper-specific attribution settings remain in their source records.

A.5 Intervention contracts and validation

Residual patching uses an RGB$(128, 128, 128)$ reference image with identical dimensions and text. Sampled exact layers are 0, 5, 10, 16, 21, 26, and 31 on SmolVLM/LLaVA; attribution estimates use all 32 layers. Visual slots and answer-prediction slots are separate intervention sets. Token sweeps select question subwords and the final four assistant-boundary subwords. The public attribution API currently replays a gradient forward per layer; within that layer one gradient covers every token. This report does not claim the optimized all-sites two-forward/one-backward cache. Four matched runs require 658 forwards and 132 activation backwards, with 18.17 GiB peak allocation.

The knockout study uses the LLaVA chat instruction "Answer the question using a single word or phrase." Its question is shown in Figure 10. Each center c jointly masks all heads in $[\max(0, c-4), \min(31, c+4)]$. Image keys, question queries, and the final prediction position retain native expanded coordinates. Its metric is $P(Y)$ for token ID 612, not full-word Yellow probability. Shirt/other masks contain 20/556 keys. Blocking connections already forbidden by the causal mask is a null control. The second case has baseline $P(Black) = 0.9982$ and a maximum image-to-question decrease of 9.43%.

Path patching uses the paired MMVP images with the same question and text layout. Sender outputs at the prediction position are replaced from the reference input; every other attention-head output is fixed to its base cache. MLP and LayerNorm recompute, the final receiver residual is cached, and a fresh base run injects only that receiver. Complete sweeps cover every declared sender head in the five models; SmolVLM, InternVL, Qwen3-VL, Pixtral, and LLaVA expose 480, 336, 448, 1,280, and 1,024 paths, respectively. Full head-output freezing is unavailable for Qwen3.5's DeltaNet stack. Selected paths are configured by the user, not automatically assigned semantic component names.

The GPT-2 EAP graph includes input, head, and MLP output messages into editable pre-LayerNorm Q/K/V branches, MLP inputs, and final readout. EAP uses the clean gradient; EAP-IG uses five left input-embedding samples $0, 0.2, 0.4, 0.6, 0.8$, preserving position embeddings and endpoint message differences. The author IOI CSV row is 475; its reference replaces the final Andrew occurrence with Jeremy. This graph experiment prepends BOS and reports its own original/reference margins, 6.1643/3.2137. It should not be numerically pooled with the separately prepared text-lens readout. Every retained-edge curve point is an actual forward with the recorded post-pruning edge count.

The VSV comparison uses BF16 LLaVA-1.5-7B, SmolVLM-500M and Qwen3-VL-2B on the original COCO validation image 310196. Its common user request is "Please help me describe the image in detail." Each model uses its native chat template. Directions are the image-bearing minus image-free last-context block residuals. The no-image input retains exactly the positive input's text IDs after removing its complete visual span and formatting. This preserves LLaVA's boundary whitespace instead of retokenizing a deleted image marker. Qwen's no-image input has explicit zero-based three-axis positions, avoiding reuse of an earlier image's RoPE offset.

All decoder block outputs are configured. Joint intervention adds the fixed direction at the original context-end and generated positions, then restores the original vector norm. The retained scales are 0, 0.005, 0.01, 0.02, 0.05, 0.10 and 0.17. Every generation uses greedy uncached full-prefix replay, a 128-token limit, and no repetition penalty. Native uncached baselines use the same full-sequence logits readout; the default HF last-position readout can change a BF16 argmax tie through matrix rounding. Smol's native parity reference retains its unmodified projected image features across every prefix; HF 5.3 otherwise removes its pixel inputs after prefill. The library generation itself replays the image at every step. Complete outputs, IDs, direction magnitudes, input contracts and zero/identity controls are archived. Figure excerpts retain at most 40 original words per cell with exact source offsets; selected visual checks are recorded separately. COCO reference captions use both skiing and snowboarding for this image, so that activity-label difference is not marked as an error.

Figure 14 reuses this LLaVA baseline's first 80 generated prefixes in every ranking condition. Its strength curves fix the prefix "The image captures a snowboarder skillfully riding"; snowboard/couch use their first pieces, rather than complete-word probabilities. The earlier FP16, manually formatted VSV runs remain historical records. The current study does not include the released implementation's PCA, normalized MLP-output rule or SLA. Repetition at large raw-residual scales characterizes this intervention configuration, rather than complete VISTA.

Same-input replacement and post-hook baseline restoration are exact in the retained shared and new paired runs. Final-site readouts differ from native BF16 logits by at most 0.125 because of selected-position matrix rounding; FP32 graph parity is within 1.26×10^{-4}. These numerical checks verify execution, not the uniqueness of the proposed interpretation. Newly added attribution/activation measurements preserve the archived input and baseline scores; plotting changes leave all source measurements unchanged.

A.6 Replaying figures and measurements

The report notebook reads bundled results and figures without weights or a GPU. All its outputs are retained. The following commands regenerate the new comparisons and build the report; dependencies include Pandoc, XeLaTeX, and Biber:

```
python examples/render_crossmodel_report.py \
  --source-dir docs/report/v10/results --output-dir docs/report/v11/lens/v2 \
  --case-ids gqa_07302654 --figure-types logit-lens
python examples/render_report_lenses_v09.py
python examples/render_report_attention_v09.py
python examples/render_report_attention_v09.py \
  --output docs/report/v11/attention/v2 --groups local --overlay-style contrast
python examples/render_report_causal_comparison.py \
  --output-dir docs/report/v11/causal/v2
python examples/render_report_vsv.py
python examples/render_causal_report.py
python examples/run_report_path.py --render-only \
  --source docs/report/v10/results/path --report-dir docs/report/v10/causal
python examples/build_technical_report.py
```

The measurement runners handle the shared matrix, attention comparison, paired residual replacements, and path sweeps. Their filenames and calls are linked in the report notebook; dedicated method notebooks retain data acquisition and inference calls. Calibrated artifacts bind model, tokenizer, activation site, and data provenance. Portable JSON and figure manifests record source hashes and exported-result equality. These artifacts support replay and new-input analysis while keeping bulk checkpoints outside the code repository.

B Additional comparisons

B.1 A second vision-classifier comparison

Figure 15 repeats the native ViT comparison on the author's elephant–zebra image with the same deletion protocol as the cat–dog control in Figure 7.

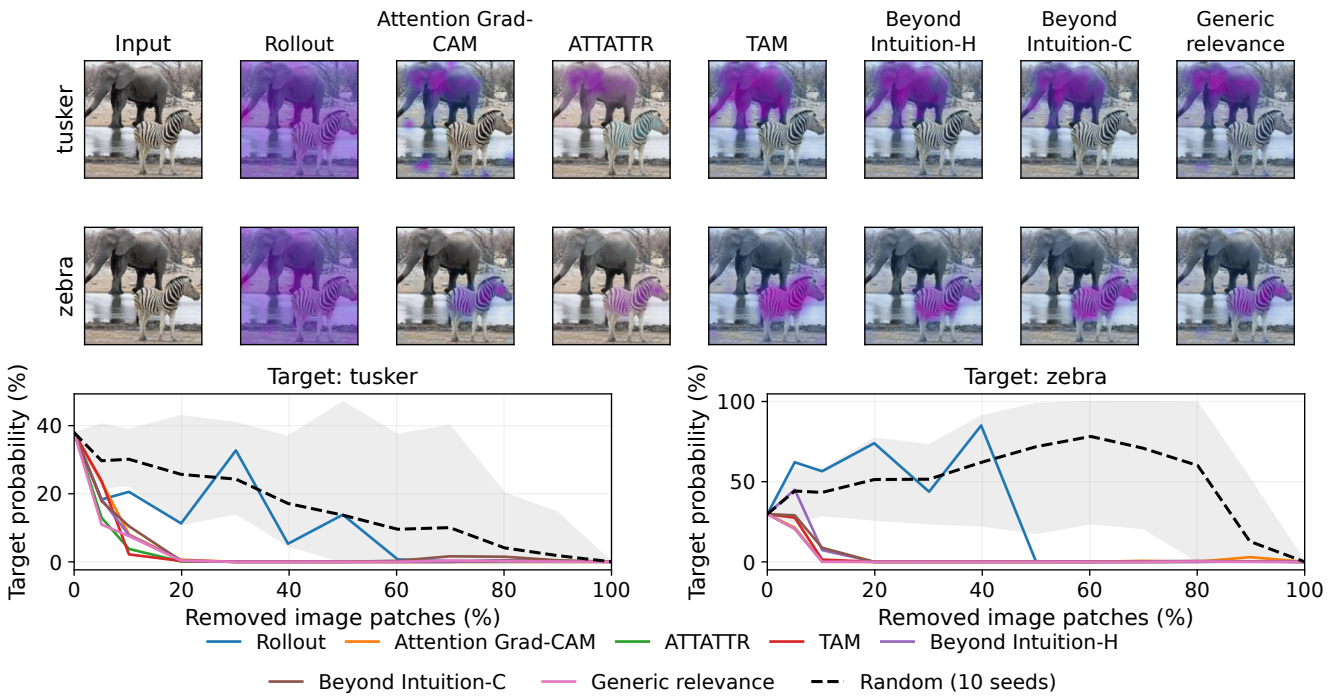


Figure 15. A second native ViT control separates elephant and zebra attribution. Identical image preprocessing and deletion budgets allow map differences to be checked against the effect of removing selected patches.

B.2 Residual replacement on the black-shirt input

Figures 16 and 17 retain the second GQA case under the same reference input, sites, and metrics as the yellow-shirt study. The smaller LLaVA effect and stronger SmolVLM question-word sensitivity illustrate why the main comparison is interpreted per case rather than as a universal pathway ranking.

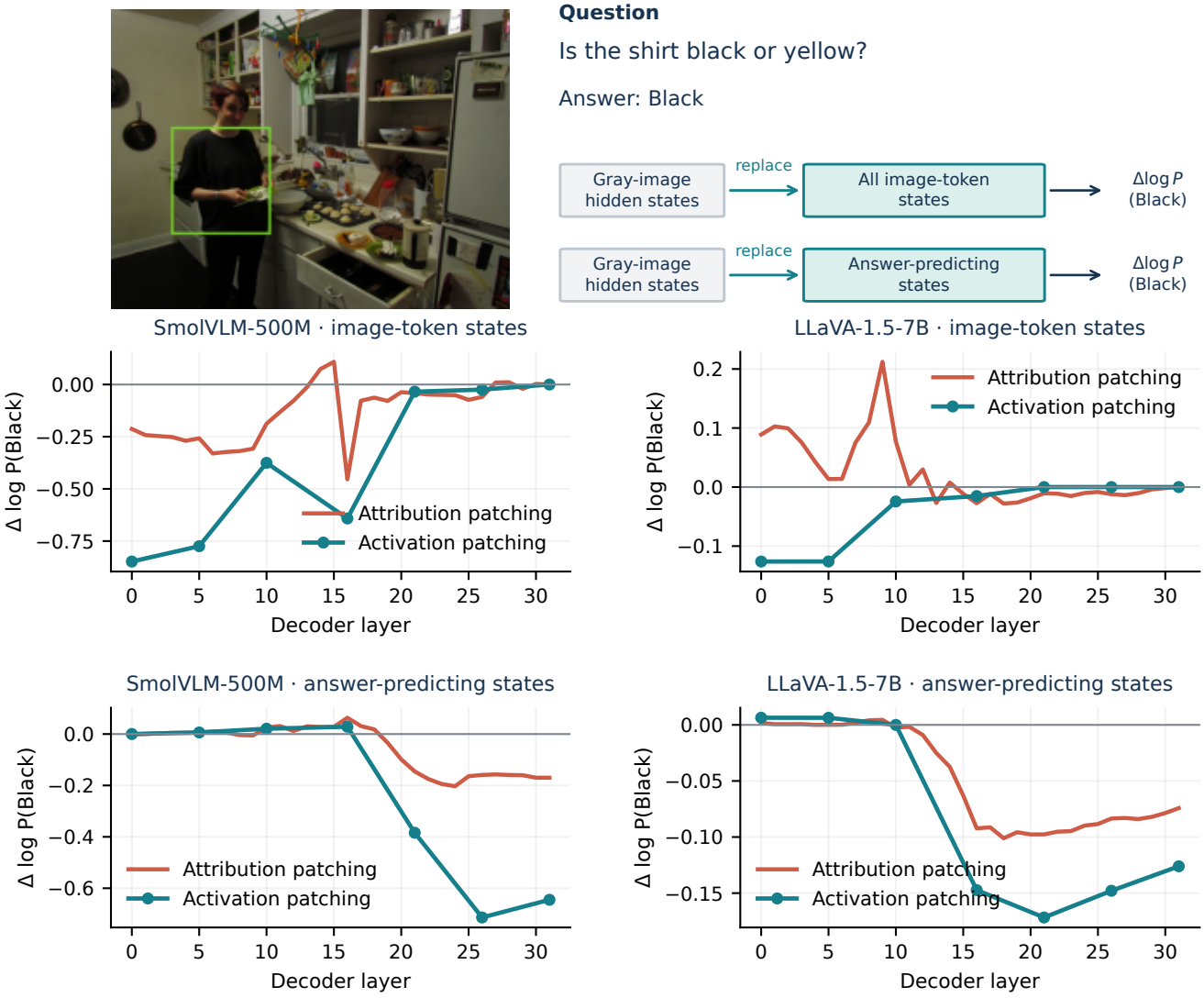


Figure 16. Exact and first-order residual replacement for the black-shirt question. LLaVA's measured effects are markedly smaller than on Yellow, whereas SmolVLM remains sensitive to the gray-image reference. Both methods use the same native sites and complete Black score.

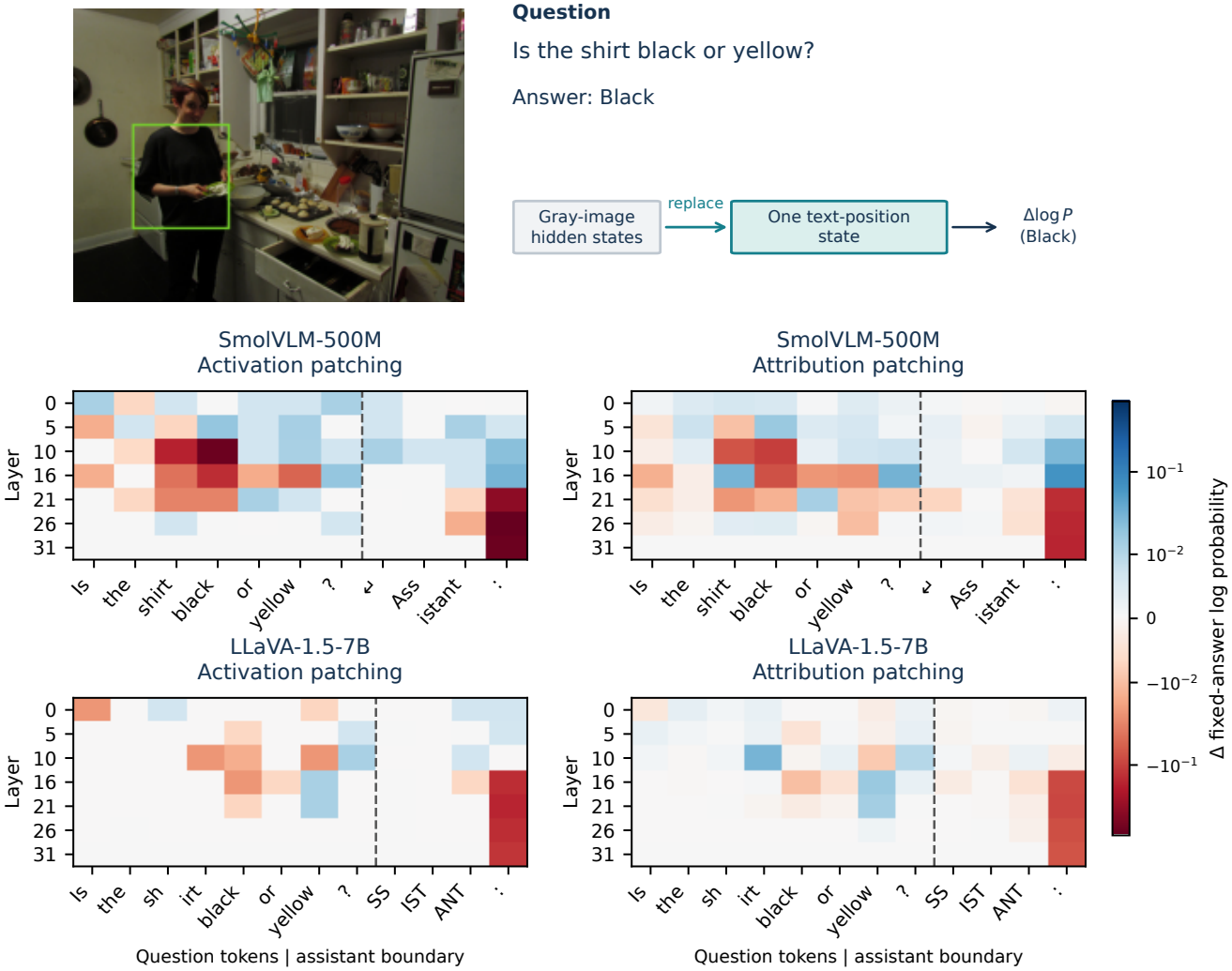


Figure 17. Text-position patching for the black-shirt input. Exact replacements identify intermediate question-word and later prediction-position effects; the aligned first-order maps provide a direct approximation check rather than a separate attribution target.